

Conjecture 00000001046 is false:

$x \mapsto x + x^{q-2}$ has differential uniformity at least 4 whenever

$$3 \mid q - 1$$

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1 The conjecture

Conjecture 00000001046 (English text, verbatim): “Conjecture: The differential uniformity of $x \mapsto x + x^{q-2}$ is 2 for large q (uniqueness of the optimal almost perfectly nonlinear (APN) function for large q).” The Chinese text says the same: the differential uniformity of $x \mapsto x + x^{q-2}$ is 2 for large q .

Reading. q is a prime power and $f_q(x) = x + x^{q-2}$ is a map $\mathbb{F}_q \rightarrow \mathbb{F}_q$. “For large q ” means: there is N such that $\delta(f_q) = 2$ for every $q > N$. We refute this in three settings, and claim no more: (R1) q ranges over all prime powers; (R2) $q = 2^m$ (the usual APN setting); (R3) q ranges over odd prime powers. The parenthetical (uniqueness of an optimal APN function) is a separate assertion; we neither use nor refute it. Only the explicit differential-uniformity clause is refuted, which already makes the conjunction false. Not refuted: a reading restricted to $q = 2^m$ with m odd, or to $q = 3^k$ (there $3 \nmid q - 1$ and our argument does not apply).

2 Definitions

Definition 1. For a finite field F and $g : F \rightarrow F$, the differential uniformity is

$$\delta(g) = \max_{a \neq 0, b \in F} \#\{x \in F : g(x+a) - g(x) = b\}.$$

This is the $c = 1$ case of the definition quoted in the source below (Yan, arXiv:2101.10543): “ $F(x)$ is differentially (c, δ) uniform, when the maximum number of solutions $x \in \text{GF}(p^n)$ of $F(x+a) - F(cx) = b$, $a, b, c \in \text{GF}(p^n)$, $c \neq 1$ if $a = 0$, is equal to δ .” With $c = 1$ the condition is $a \neq 0$. In characteristic 2, $-$ equals $+$.

3 Proof

Lemma 2. If $q = |F| \geq 3$ then $x^{q-2} = x^{-1}$ for $x \neq 0$ and $0^{q-2} = 0$. Hence $f(x) = x + x^{-1}$, with the convention $0^{-1} = 0$.

Proof. For $x \neq 0$, $x^{q-1} = 1$ (Lagrange in $F^\times$), so $x^{q-2} \cdot x = 1$. Since $q - 2 \geq 1$, $0^{q-2} = 0$. □

Lemma 3. If $3 \mid q - 1$ there is $\omega \in F$ with $\omega \neq 1$ and $\omega^3 = 1$; then $\omega^2 + \omega + 1 = 0$.

Proof. $|F^\times| = q - 1$ is divisible by the prime 3, so by Cauchy's theorem $F^\times$ has an element of order 3. From $(\omega - 1)(\omega^2 + \omega + 1) = \omega^3 - 1 = 0$ and $\omega \neq 1$ we get $\omega^2 + \omega + 1 = 0$. $\square$

Theorem 4. *If $3 \mid q - 1$ then the equation $f(x + 1) - f(x) = 2$ has the four distinct solutions $x = 0, -1, \omega, \omega^2$. Hence $\delta(f_q) \geq 4$, so $\delta(f_q) \neq 2$.*

Proof. $q - 1 \geq 3$, so $q \geq 4$ and Lemma 2 applies. We have $\omega^{-1} = \omega^2$, $(\omega^2)^{-1} = \omega$, $\omega + 1 = -\omega^2$ and $\omega^2 + 1 = -\omega$.

- $x = 0$: $f(1) - f(0) = (1 + 1) - 0 = 2$.
- $x = -1$: $f(0) - f(-1) = 0 - (-1 - 1) = 2$.
- $x = \omega$: $f(\omega + 1) - f(\omega) = (-\omega^2 - \omega) - (\omega + \omega^2) = -2(\omega^2 + \omega) = 2$.
- $x = \omega^2$: $f(\omega^2 + 1) - f(\omega^2) = (-\omega - \omega^2) - (\omega^2 + \omega) = 2$.

Distinctness: $\omega \neq 0$; $\omega \neq -1$ (else $\omega^2 + \omega + 1 = 1$); $\omega^2 \neq -1$ (else $\omega = \omega^2 + \omega + 1 - \omega^2 - 1 = 0$); $\omega^2 \neq \omega$ (else $\omega = 1$); $0 \neq -1$; $\omega^2 \neq 0$. Taking $a = 1 \neq 0$, $b = 2$ gives $\delta(f_q) \geq 4$. $\square$

Theorem 5. *(R1) For every N there is a finite field with $q > N$ and $\delta(f_q) \geq 4$. (R2) For every $k \geq 1$, $\delta(f_{4^k}) \geq 4$; so no N has $\delta(f_{2^m}) = 2$ for all $m \geq N$. (R3) For every $k \geq 1$, $\delta(f_{7^k}) \geq 4$; so no N has $\delta(f_q) = 2$ for all odd $q > N$.*

Proof. $4 \equiv 7 \equiv 1 \pmod{3}$, so $3 \mid 4^k - 1$ and $3 \mid 7^k - 1$; apply Theorem 4 to $\mathbb{F}_{4^k} = \text{GaloisField}(2, 2k)$ and $\mathbb{F}_{7^k} = \text{GaloisField}(7, k)$, which have 2^{2k} and 7^k elements and are arbitrarily large. $\square$

Numerical check (not used in the proof): $\delta(f_{2^m}) = 4$ for $m = 2, 4, 6, 8$ and $= 2$ for $m = 3, 5, 7$.

4 Lean formalization

File `lean/Conjecture1046/Basic.lean` (221 lines, Lean 4 + Mathlib, only import `Mathlib`).
Definitions: `f x = x + x ^ (Fintype.card F - 2)`; `diffCount g a b = #\{x \mid g(x+a) - g(x) = b\}`; `diffUniformity g` is the `Finset.sup` of `diffCount g a b` over $a \neq 0$ and all b . Main theorems:

- `four_le_diffUniformity`: for any finite field F with $3 \mid \text{Fintype.card } F - 1$, $4 \leq \text{diffUniformity } f$.
- `not_eventually_two`: no N such that every finite field F with $N < |F|$ has `diffUniformity f = 2` (R1).
- `not_eventually_two_char_two`: no N such that `diffUniformity f = 2` on `GaloisField 2 m` for all $m \geq N$, $m \neq 0$ (R2); via `char_two_even`.
- `not_eventually_two_odd`: the same over all finite fields of odd order larger than N (R3); via `char_seven`.

Supporting lemmas: `pow_card_sub_two` (Lemma 2), `exists_cube_root` (Lemma 3, from Mathlib's `exists_prime_orderOf_dvd_card`), `four_le_diffCount` (Theorem 4). `#print axioms` for each main theorem gives `[propext, Classical.choice, Quot.sound]`; there is no `sorry` and no `native_decide`. Build: `lake build` with Mathlib v4.33.1.

5 Relation to the earlier submission

An earlier submission (PR #157, orionsheep) was closed. Its Lean checked only three instances $q = 16, 256, 1024$, and the reviewer wrote: “Three data points are consistent with $\delta = 2$ for all $q > 1024$, so the asymptotic claim is untouched. The genuine argument [...] appears only in prose. Happy to re-review a version proving the general even- m case in Lean.” This submission proves the general case in Lean: $\delta \geq 4$ for every finite field with $3 \mid q - 1$, which includes $\mathbb{F}_{2^m}$ for every even $m \geq 2$ and $\mathbb{F}_{7^k}$ for every $k \geq 1$, and it states the refutation of the eventual claim directly.

Source

H. Yan, “On -1 -differential uniformity of ternary APN power functions”, arXiv:2101.10543, <https://arxiv.org/abs/2101.10543> (abstract; definition quoted above).